9 Fisheries Impact

9.1 Introduction

9.1.1.1 This Section of EIA presents the approach to and the findings of the fisheries impact assessment based on the RODP shown in **Figure 3.1**.

9.1.1.2 The fisheries impact assessment has been prepared in accordance with the EIA Study Brief No. ESB-246/2012 for the proposed Project "Housing Sites in Yuen Long South" (hereinafter referred to as "the Project").

9.1.1.3 The Project for the purpose of EIA refers to the planning and engineering feasibility study to review the development potential of the Potential Development Area (PDA) in Yuen Long South with a view to identifying housing sites for private and public housing developments and the associated infrastructural works for the proposed developments. PDA refers to the areas within the boundary of the proposed development design in Yuen Long South, including the public and private housing developments and the government, institution or community facilities and open space and/or amenities.

9.1.1.4 The Assessment Area for fisheries impact assessment as stipulated under Section 3.4.6 of the EIA Study Brief No. ESB-246/2012 shall include all areas within 500m distance from the site boundaries of the Project and shall be extended to include other areas if they are to be impacted by the construction and operation of the Project during the course of the EIA Study. Special attention shall be paid to pond culture resources and activities as well as any watercourses which serve as water sources for fish ponds.

9.1.1.5 The fisheries impact assessment has been conducted in accordance with requirement of Annexes 9 and 17 of TM-EIAO as well as the requirements set out under Section 3.4.6 and Appendix 7.7 of the EIA Study Brief.

9.2 Legislation, Standards and Guidelines

9.2.1.1 The relevant legislation, standards and guidelines applicable to the present study for the assessment of fisheries impact include:

- Environmental Impact Assessment Ordinance (EIA Ordinance) (Cap. 499)
- Water Pollution Ordinance (Cap. 358)
- Fisheries Protection Ordinance (Cap. 171)
- Marine Fish Culture Ordinance (Cap. 353)
- The general criteria and guidelines for evaluation and assessing fisheries impacts and considering mitigation measures are documented in the Annexes 9 and 17 of the Technical Memorandum on Environmental Impact Assessment Process (TM-EIAO)

9.3 Methodology for Baseline Establishment

9.3.1 Literature Review

9.3.1.1 A desktop study was conducted to assess the baseline conditions of pond fish culture activities and resources within the Assessment Area with respect to the fisheries activities of Hong Kong as a whole. Information and literature reviewed include:

- AFCD Annual Reports (1997 – 2016);
- AFCD website

(http://www.afcd.gov.hk/english/fisheries/fish_aqu/fish_aqu.html)

- AFCD Accredited Fish Farm Scheme Website

(http://www.hkaffs.org/en/index.html)

- EIA & TIA Studies for the Stage 2 of PWP Item No. 215DS-Yuen Long and Kam Tin Sewerage and Sewage Disposal (YLKTSSD) prepared by Ove Arup & Partners Hong Kong Ltd (2004) (Ref. No. EIA-094/2004)
- EIA for Hung Shui Kiu New Development Area prepared by AECOM (2016) (Ref. No. EIA-248/2016)
- AFCD's internal records of ponds

9.3.2 Site Survey

9.3.2.1 Site surveys were initially conducted during February 2013 and May 2013 to investigate the actual status of fisheries and the conditions of fisheries resources within the Assessment Area. Additional surveys were conducted during March 2014 – April 2014 and during September 2016 – May 2017 to verify the conditions of the ponds identified at later stages as a result of the expansion of Assessment Area of the RODP. All ponds within the Assessment Area were identified and mapped on plan. Fisheries operation such as stocking, feeding and harvesting, the presence of equipment or devices related to fisheries operation, and signs of maintenance of fish pond bunds were recorded. Observations were also supplemented by any fisheries-related activities observed during other field surveys for terrestrial ecology. Photographs of the identified fish ponds were taken during the surveys (see **Appendix 9.1**).

9.3.2.2 Based upon observations during site surveys, ponds within the Assessment Area were categorised as follows:

- **Active**: ponds currently utilised for commercial aquaculture activities; this category includes commercial fish ponds, fish fry ponds and water flea ponds;
- **Inactive**: no current commercial aquaculture activities, but no major physical constraints to the resumption of such activity in the short-term; this includes ponds with fish present in non-commercial quantities and ponds used for casual sport fishing or for water sports;
- **Abandoned**: ponds in which there is physical evidence that aquaculture has not taken place for many years (e.g. ponds overgrown with vegetation) and/or where there are obvious physical constraints to the resumption of fisheries activities (e.g. ponds that are fenced off and

inaccessible); ponds that are being utilised for non-fisheries related activities and concreted ornamental/ landscaped ponds are also included in this category.

9.3.2.3 During the site surveys, interviews with local villagers or fish farmers were conducted wherever possible to obtain information about fisheries activities in the Assessment Area.

9.4 Description of Existing Fisheries Baseline Conditions

9.4.1 Review of Existing Information

9.4.1.1 Pond fish culture has been centred in the north-west New Territories for decades. Currently the majority (~96% in 2016) of local pond fish culture belong to polyculture, with Bighead Carp *Aristichthys nobilis*, Grass Carp *Ctenopharyngodon idellus*, Common Carp *Cyprinus carpio* and Silver Carp *Hypophthalmichthys molitrix* in a combination with Tilapia *Oreochromis* spp. or Grey Mullet *Chelon labrosus* being the most dominant culture species. Approximately 4% of local pond fish culture belongs to monoculture of carnivorous species such as Giant Groups *Epinephelus lanceolatus*, Seabreams *Acanthopagrus* spp. and Spotted Scat *Scatophagus argus* in brackish fish ponds near to the coastline. Most of the fry and fingerlings are imported from the Mainland and Taiwan, whilst some of the Grey Mullet fry may also be captured from local coastal waters.

9.4.1.2 In order to help local fish farms to develop, AFCD has launched the "Accredited Fish Farm Scheme" (AFFS). Under this scheme, products from registered fish farms are "accredited" by AFCD, and can be marketed under the unique brand name of the scheme. Fish farms registered under the scheme will be regularly inspected by AFCD officers, checking on the farm hygiene conditions, the maintenance of management records, water quality and fish health conditions. A series of advisory leaflets and guidelines on aquaculture management, including "Good Aquaculture Practices Series 3 – Environmental Management on Pond Fish Culture" and "Good Aquaculture Practices Series 5 – Fry Health Management" have been published by Aquaculture Fisheries Division of AFCD. These publications provide guidelines on physical environment requirements for establishment of fish pond farms, drainage requirements, water quality, and a list of good fish pond culture practices for pond fisheries and fry management.

9.4.1.3 According to information from AFCD website, local inland ponds cover an area of approximately 1,135ha, producing 2,092 tonnes of freshwater fish amounting to $53million in 2016. **Table 9.1** summarises the annual pond fish production and fish pond area between 1997 and 2016 based on information extracted from AFCD annual report and website.

Table 9.1 Annual Pond Fish Production and Fish Pond Area

Year	Pond Fish Production (tonnes)	Fish Pond Area (ha)	Overall Pond Fish Production Rate (kg/ha/year)
1997	5000	1125	4444
1998	4900	1110	4414
1999	4500	1094	4113
2000	2817	1060	2657
2001	2550	1059	2407

Year	Pond Fish Production (tonnes)	Fish Pond Area (ha)	Overall Pond Fish Production Rate (kg/ha/year)
2002	1989	1030	1931
2003	2114	1029	2054
2004	1977	1026	1927
2005	1897	1026	1849
2006	1943	1024	1897
2007	1927	1160	1661
2008	2266	1160	1953
2009	2105	1160	1814
2010	2190	1109	1975
2011	2315	1130	2049
2012	2306	1150	2005
2013	2187	1150	1902
2014	2001	1140	1755
2015	2092	1140	1835
2016	2543	1135	2241

9.4.2 Field Surveys

9.4.2.1 A total of 51 ponds were identified within or close to the Assessment Area as listed in **Table 9.2** below and their locations and status are displayed in **Figure 9.1**. Photographs of each identified ponds taken during site surveys are presented in **Appendix 9.1**. The locations and extent of ponds were mapped on plan in **Figure 9.1** based on the information from GeoInfo Map from Lands Department and further verified through updated field surveys.

9.4.2.2 A total of nine ponds identified within PDA, including five shallow farm ponds (Ponds #5 - #9) near Sha Tseng Tsuen in Area 1, a small, managed shallow pond (Pond #17) to the east of Lam Tai East Road in Area 2, two abandoned and overgrown ponds (Ponds #23 and #24) located near the southern end of Kung Kum Road, and an abandoned and dry pond (Pond #37) located near the southern end of Long Hon Road in Area 1 of the PDA. All the farm ponds (Ponds #5 – 9) in Area 1 within the PDA are shallow farm ponds being used for irrigation. In the most recent survey in 2016, Pond #17 was found containing very shallow water (not suitable for fisheries purposes) and partially fenced. The pond appeared to be actively managed but the use of this pond was uncertain. There are some other farm ponds (Ponds #1, #10, #12, #13 and #42) located in the Assessment Area but outside of the Project Area. These farm ponds are either ponds for retention of rainwater for irrigation or inactive wet-fields for wet agriculture.

9.4.2.3 There are a total of four active fish ponds identified during the surveys, including two small fish ponds/ managed ponds (Ponds #3 and #4) located behind the dense woodland behind Shan Ha Tsuen and outside of the PDA. These two ponds were being operated by the local villagers for cultivation of Nile Tilipia *Oreochromis niloticus*. However, it is not certain if the fish are commercial fish stocks or for self-consumption by the ponds' owner(s). The remaining active fish ponds (Pond #34 and #35) are located at about 700m to the Southeast of the southern end of Kiu Hing Road and are located outside of the 500m Assessment Area. Fisheries-related equipment was seen at nearby fishponds #34 and #35. However, the scale of operations of these fishponds is small because of their small sizes and isolated locations, making their nature and commercial uses uncertain.

9.4.2.4 There is an inactive fish pond (Pond #22) recorded near Fui Sha Wai outside of the Project area.

9.4.2.5 There are some ponds which are being actively disturbed/ managed for non-fisheries related uses, including two Fung Shui Ponds/ Landscaped Ponds (Ponds #2 and #21) of the local villages, seven stormwater ponds (Ponds #14, #16, #39, and #48 - #51) which are largely concreted, four mitigation ponds (Ponds #43, #44, #46 and #47), and some ponds which have recently been filled (Ponds #20, #28 - #32, and #38). The remaining ponds identified during surveys belong to abandoned ponds heavily overgrown with vegetation.

Table 9.2 Status of Ponds within Assessment Area

Pond Ref. No.	Location	Within PDA/ Project Area (Yes/ No)	Status	Area (ha)	Remarks
1	Near Yeung Ka Tsuen	No	Abandoned	0.18	Appeared to be a former farm pond for irrigation
2	Near Fui Sha Wai	No	Abandoned	0.16	Fung Shui Pond/ Landscaped Pond
3	~250m west of southern end of Long Hon Road	No	Possibly Active	0.10	Actively managed pond for cultivation of Tilapia; commercial use is uncertain
4	~250m west of southern end of Long Hon Road	No	Possibly Active	0.02	Actively managed pond for cultivation of Tilapia; commercial use is uncertain
5	Near Sha Tseng Tsuen	Yes	Abandoned	0.03	Active farm pond (shallow pond) for irrigation
6	Near Sha Tseng Tsuen	Yes	Abandoned	0.03	Active farm pond (shallow pond) for irrigation
7	Near Sha Tseng Tsuen	Yes	Abandoned	0.18	Active farm pond (shallow pond) for irrigation
8	Near Sha Tseng Tsuen	Yes	Abandoned	0.10	Active farm pond (shallow pond) for irrigation
9	Near Sha Tseng Tsuen	Yes	Abandoned	0.03	Active farm pond for irrigation
10	South of Shan Ha Tsuen	No	Abandoned	0.12	Appeared to be a former farm pond for irrigation
11	Within woodland adjacent to Tan Kwai Tsuen Fresh Water Service Reservoir	No	Abandoned	0.03	Abandoned pond within woodland
12	Near Fui Sha Wai	No	Abandoned	0.12	Active farm pond (shallow pond) for irrigation
13	Near Yeung Uk Tsuen	No	Abandoned	0.21	Appeared to be a former farm pond for irrigation
14	At Kiu Tau Wai	No	Abandoned	0.43	Stormwater pond in Kiu Tau Wai Stormwater Pumping Station
15	Near Sheung Cheung Wai	No	Abandoned	0.60	Marsh Pond
16	Near Sheung Cheung Wai	No	Abandoned	0.63	Stormwater pond in Sheung Cheung Wai Stormwater Pumping Station

Pond Ref. No.	Location	Within PDA/ Project Area (Yes/ No)	Status	Area (ha)	Remarks
17	~80m east of middle of Lam Tai East Road	Yes	Abandoned	0.04	Managed pond (shallow pond); nature and commercial use uncertain
18	Near Wong Nai Tun Tsuen	No	Abandoned	0.05	Overgrown pond next to a house
19	Near Tai Tong Tsuen	No	Abandoned	0.17	Overgrown pond
20	Near Yeung Ka Tsuen	No	Abandoned & Filled	0.31	Filled
21	Near Ying Lung Wai	No	Abandoned	0.09	Fung Shui Pond/ Landscaped Pond
22	Near Fui Sha Wai	No	Inactive	0.25	Inactive Fish Pond near Fui Sha Wai
23	~300m near the southern end of Kung Um Road	Yes	Abandoned	0.12	Overgrown with vegetation
24	~300m near the southern end of Kung Um Road	Yes	Abandoned	0.05	Overgrown with vegetation
25	Near Kong Tau Tsuen	No	Abandoned	0.03	Overgrown with vegetation
26	Near Kong Tau Tsuen	No	Abandoned	0.02	Overgrown with vegetation
27	Near Tai Kei Leng	No	Abandoned	0.03	Overgrown with vegetation
28	Near Tai Tong Tsuen	No	Abandoned & Filled	0.05	Filled
29	Near Tai Tong Tsuen	No	Abandoned & Filled	0.11	Filled
30	Near Tai Tong Tsuen	No	Abandoned & Filled	0.02	Filled
31	Near Tai Tong Tsuen	No	Abandoned & Filled	0.05	Filled
32	Near Tai Tong Tsuen	No	Abandoned & Filled	0.05	Filled
33	~700m near the southern end of Kiu Hing Road	No	Abandoned	0.04	Overgrown with vegetation
34	~700m near the southern end of Kiu Hing Road	No	Possibly Active	0.03	Managed fish pond; commercial use uncertain
35	~700m near the southern end of Kiu Hing Road	No	Possibly Active	0.05	Managed fish pond; commercial use uncertain
36	Near Yeung Ka Tsuen	No	Abandoned	0.03	Overgrown with vegetation
37	Near the southern end of Long Hon Road	Yes	Abandoned	0.10	Dried out and overgrown with vegetation
38	Near Tai Tong Tsuen	No	Abandoned & Filled	0.01	Filled
39	At Ma Tin Tsuen	No	Abandoned	0.12	Stormwater pond in Ma Tin Tsuen Stormwater Pumping Station
40	~600m northwest of	No	Abandoned	0.13	Landscaped Pond

Pond Ref. No.	Location	Within PDA/ Project Area (Yes/ No)	Status	Area (ha)	Remarks
	San Sang San Tsuen				
41	~600m northwest of San Sang San Tsuen	No	Abandoned	0.31	Overgrown with vegetation
42	~600m northwest of San Sang San Tsuen	No	Abandoned	0.21	Active farm pond for irrigation
43	~500m north of San Sang San Tsuen under Western Corridor	No	Abandoned	0.07	Mitigation pond
44	~500m north of San Sang San Tsuen under Western Corridor	No	Abandoned	0.08	Mitigation pond
45	~500m north of San Sang San Tsuen under Western Corridor	No	Abandoned	0.09	Overgrown with vegetation
46	~500m north of San Sang San Tsuen under Western Corridor	No	Abandoned	0.18	Mitigation pond
47	~500m north of San Sang San Tsuen under Western Corridor	No	Abandoned	0.13	Mitigation pond
48	At Sik Kong Wai	No	Abandoned	0.34	Stormwater pond in Sik Kong Wai Stormwater Pumping Station
49	At Sik Kong Wai	No	Abandoned	0.31	Stormwater pond in Sik Kong Wai Stormwater Pumping Station
50	At Sik Kong Tsuen	No	Abandoned	0.18	Stormwater pond in Sik Kong Tsuen
51	At Lo Uk Tsuen	No	Abandoned	0.32	Stormwater pond next to Lo Uk Tsuen

9.5 Identification and Evaluation of Impacts

9.5.1 Construction Phase – Direct Impacts

9.5.1.1 Potential direct impacts on pond fisheries during construction phase could be resulted from temporary or permanent loss of fish ponds or areas for fish pond operation during construction of the Project. Of the nine ponds identified within the PDA, a total of eight ponds (Ponds #5 - #9, #17, #23 - #24) will be permanently lost due to the construction of the Project. The five ponds (Ponds #5 - #9) identified in the Area 1 of PDA are active farm ponds (near Sha Tseng Tsuen) which are all shallow ponds for retention of water for irrigation purposes (or inactive wet-fields for wet agriculture). The small, shallow pond located to east of Lam Tai East Road in the Area 2 (#17) appeared to be actively managed but it was too shallow for fisheries operation and the use of the pond was uncertain. There are two overgrown and abandoned ponds (Ponds #23 and #24) in Area 3 which will be permanently lost due to construction of the Project. The abandoned and dry pond (i.e. Pond #37) located in Area 1 of PDA will be preserved in Green Belt (GB) zone in the RODP. All the eight ponds namely Ponds #5 - #9, #17 and #23 - #24

are considered as "Abandoned Fish Ponds" as they have no historical records of fisheries-related uses. No active fish ponds will be lost due to the Project. As the affected 'abandoned fishponds' are all small in size (ranging from 0.03ha to 0.18ha and approximately 0.58ha in total), the direct fisheries impacts on pond fisheries/ aquaculture are considered to be negligible when taking into account the potential that these ponds are all resumed as active fish ponds for commercial fisheries operation.

9.5.2 Construction Phase – Indirect Impacts

9.5.2.1 Potential sources for construction phase indirect impacts on pond fisheries include impacts on bund stability, water seepage, blockage of access to fish ponds, disruption of hydrology and deterioration of water quality or degradation of the environmental conditions of the fish ponds.

9.5.2.2 The only possibly active fish ponds that will be retained within 500m Assessment Area are Ponds #3 and #4 located approximately 250m west of the southern end of Long Hon Road, i.e. approximately 150m from the nearest boundary of Area 1 of PDA. Given the long buffer distances of these two possibly active fish ponds from the Project Area which is located at the downstream, no significant indirect impacts arising from construction phase of the Project resulting from degradation or deterioration of these two active fish ponds are anticipated.

9.5.2.3 The remaining ponds identified within 500m Assessment Area comprise mainly abandoned and overgrown ponds, and an inactive fish pond. With good site practices and proper control of construction site run off, sewage discharge from work force, other sources of construction waste waters and accidental spillage as detailed in **Section 6**, the induced water quality impact as a result of deterioration of water quality or degradation of the environmental conditions of fish ponds is considered to be low.

9.5.3 Operational Phase – Direct Impacts

9.5.3.1 The Project will result in permanent loss of eight abandoned ponds with no signs of fisheries-related uses (Ponds #5 – 9, #17, #23 and #24) within the PDA. The impact of loss of these ponds has been evaluated under **Section 9.5.1**. No other direct impacts during operational phase of the Project are anticipated.

9.5.4 Operational Phase – Indirect Impacts

9.5.4.1 Potential sources of indirect impacts on pond fisheries during operational phase of the Project may include blockage of access to active fishponds, and degradation/ deterioration of environmental conditions of fish ponds due to induced-water quality impacts from sewage and runoff from the proposed development. As the only two possibly active fish ponds (Ponds #3 and #4) that will be retained within the 500m Assessment Area are located in upstream area and at long buffer distances from the Project Area, no significant indirect impacts resulting from degradation of the environmental conditions of or blockage of access to these active fish ponds are anticipated during operation of the Project. In addition, the Project will provide an sewage treatment works and associated sewerage facilities to serve the proposed development. It is expected that during the operational phase of the Project, sewage and domestic discharge within the PDA will be properly collected and treated as detailed in **Sections 6** and **7**. As such, no significant

indirect impacts as a result of degradation of the environmental conditions of existing or potential fish ponds (i.e. inactive/ abandoned ponds that could be resumed for fisheries operation) are anticipated during the operational phase of the Project.

9.5.5 Cumulative Impacts

9.5.5.1 As the Project is not expected to result in any significant direct or indirect impacts on pond fisheries during construction and operational phases of the Project, no significant cumulative impacts arising from concurrent projects on pond fisheries are anticipated.

9.6 Mitigation Measures

9.6.1.1 As no significant impacts arising from construction and operational phases of the Project on pond fisheries are anticipated, no mitigation measures on fisheries are required. However, the good site practices for the control of construction site runoff as detailed in **Section 6** shall be fully implemented to minimise impacts on water sources for potential sites for pond fish culture in the Project Area or its vicinity.

9.7 Residual Impacts

9.7.1.1 With proper good site practices implemented during construction phase, no residual impacts on pond fish culture are anticipated.

9.8 Environmental Monitoring and Audit Programme

9.8.1.1 No significant fisheries impacts are anticipated from the construction phase and operation phase of the Project. Hence, the formulation and the implementation of an environmental monitoring and audit programme for assessing the impacts on fisheries resources and fisheries operation is not considered necessary. However, good site management during construction phase as mentioned in **Section 6** will be implemented so that the treated runoff will be discharged to public drainage system in compliance with the WPCO.

9.9 Conclusion

9.9.1.1 A comprehensive review of the status of pond fish culture has been conducted through desktop study and field surveys to establish the fisheries baseline of the Assessment Area, with the assessment of fisheries conducted in accordance with the TM-EIAO requirements.

9.9.1.2 Of the total 51 ponds identified in the Assessment Area, only nine ponds were identified within PDA. These include five shallow farm ponds for irrigation (Ponds #5 - #9) near Sha Tseng Tsuen in Area 1, a small, shallow pond appeared to be actively managed with uncertain use (Pond #17) located to near the southern end of Lam Tai East Road in Area 2, two abandoned and overgrown ponds (Ponds #23 and #24) located near the southern end of Kung Kum Road, and an abandoned and dry pond (Pond #37) located near the southern end of Long Hon Road in Area 1 of the PDA. Of these nine ponds identified in PDA, only the abandoned and dry pond (Pond #37) in Area 1 will be preserved in the proposed Green Belt (GB) zone in the RODP. The remaining eight ponds will be lost due to the Project. As all the eight ponds to be lost within the PDA serving non-fisheries related purposes (of

approximately 0.58ha in total) are small in overall area, the direct fisheries impacts on pond fisheries/ aquaculture are considered to be negligible when taking into account the potential that these ponds are all resumed as active fish ponds for commercial fisheries operation.

9.9.1.3 With the separation distances (~150m) from the nearest boundary of the PDA, no direct or indirect impacts on the two possibly active fishponds (Ponds #3 and #4) are anticipated during the construction and operational phases of the Project. As such, no fisheries mitigation measures are considered necessary. In addition, the Project will provide a sewage treatment works and associated sewerage facilities to serve the proposed development. It is expected that during the operational phase of the Project, sewage and domestic discharge within the PDA will be properly collected and treated as detailed in **Sections 6** and **7**. As such, no significant indirect impacts as a result of degradation of the environmental conditions of existing or potential fish ponds (i.e. inactive / abandoned ponds that could be resumed for fisheries operation) are anticipated during the operational phase of the Project.

9.9.1.4 With the full implementation of the good site practices to control sewage discharge and site runoff as detailed in **Section 6**, no adverse induced water quality impacts on water sources for potential fish pond sites are anticipated.

9.10 References

AFCD Annual Reports (1997 – 2016) (http://www.afcd.gov.hk/english/publications/publications_dep/publications_dep.html) visited in May 2017.

AFCD website (http://www.afcd.gov.hk/english/fisheries/fish_aqu/fish_aqu.html) visited in May 2017.

AFCD Accredited Fish Farm Scheme Website (http://www.hkaffs.org/en/index.html) downloaded on 18th February 2014.